

**Program Schemes for BE B.E. CSE in Artificial Intelligence (Microsoft)**
**Cluster:** Engineering

**Institute:** UIE

**Department:** AIT-CSE

**Type of Program:** UG

| First Year  |                                                 |                   |         | Second Year |                                                   |                     |         |
|-------------|-------------------------------------------------|-------------------|---------|-------------|---------------------------------------------------|---------------------|---------|
|             | Course Name                                     | Course Category   | Credits |             | Course Name                                       | Course Category     | Credits |
| Semester -1 | Mathematics-I                                   | Program Core      | 5       | Semester-3  | Object-Oriented Programming using Java            | Program Core        | 4       |
|             | Logical Thinking & Problem Solving (C Language) | Program Core      | 4       |             | Computer Organization and Architecture            | Program Core        | 3       |
|             | Engineering Physics                             | Program Core      | 4       |             | Operating System                                  | Program Core        | 4       |
|             | Academic & Career Pathway Planning              | Value Education   | 0       |             | Data Structures and Algorithms-II                 | Program Core        | 4       |
|             | Semiconductor Physics and Applications          | Program Core      | 3       |             | Exploratory Data Analysis                         | Program Elective    | 3       |
|             | Cybersecurity Fundamentals                      | Program Core      | 1       |             | Universal Human Values, Ethics and Life Skills-II | Value Education     | 2       |
|             | Communication Skills-I                          | Skill Enhancement | 1       |             | Social Internship                                 | Internship          | 2       |
|             | Design Thinking                                 | Skill Enhancement | 1       |             | Soft Skills-I                                     | Skill Enhancement   | 1       |
|             | Cloud Fundamentals                              | Certification     | 1       |             | Entrepreneurship Mind-set                         | Ability Enhancement | 1       |
|             |                                                 |                   |         |             | Critical Enquiry and Research Methodology         | Ability Enhancement | 1       |
|             |                                                 |                   |         |             | DP-900                                            | Certification       | 1       |
|             |                                                 |                   |         |             |                                                   |                     |         |
|             | Course Name                                     | Course Category   | Credits |             | Course Name                                       | Course Category     | Credits |

|                    |                                                  |                     |   |                   |                                                |                        |                |
|--------------------|--------------------------------------------------|---------------------|---|-------------------|------------------------------------------------|------------------------|----------------|
| <b>Semester -2</b> | Mathematics-II                                   | Program Core        | 5 | <b>Semester-4</b> | Applied Machine Learning                       | Program Core           | 3              |
|                    | Data Structures and Algorithms-I                 | Program Core        | 4 |                   | Database Management System                     | Program Core           | 3              |
|                    | Digital Electronics                              | Program Core        | 4 |                   | Generative AI Foundations & Prompt Engineering | Program Core           | 4              |
|                    | Communication Skills-II                          | Skill Enhancement   | 1 |                   | Backend Development using FastAPI              | Program Core           | 2              |
|                    | General Proficiency-I                            | Ability Enhancement | 1 |                   | Computer Networks                              | Program Core           | 3              |
|                    | Universal Human Values, Ethics and Life Skills-I | Value Education     | 1 |                   | Frontend Development using React               | Program Core           | 3              |
|                    | Aptitude-I                                       | Ability Enhancement | 1 |                   | AI-102                                         | Certification          | 1              |
|                    | Computer Graphics Lab                            | Program Core        | 1 |                   | <b>RIE-1 (Semester-4)</b>                      |                        |                |
|                    | Biology for Engineers                            | Program Core        | 3 |                   | <b>Course Name</b>                             | <b>Course Category</b> | <b>Credits</b> |
|                    | AI Fundamentals                                  | Certification       | 1 |                   | Employability - Aptitude-II                    | Skill Enhancement      | 1              |
|                    |                                                  |                     |   |                   | Innovation - Marketing & Management Essentials | Skill Enhancement      | 1              |
|                    |                                                  |                     |   |                   | Research - Academic Writing                    | Skill Enhancement      | 1              |
|                    |                                                  |                     |   |                   | <b>RIE-2 (Semester-4)</b>                      |                        |                |
|                    |                                                  |                     |   |                   | <b>Course Name</b>                             | <b>Course Category</b> | <b>Credits</b> |
|                    |                                                  |                     |   |                   |                                                |                        |                |
|                    |                                                  |                     |   |                   | Employability - Soft Skills-II                 | Ability Enhancement    | 1              |

|  |  |  |  |                                                                          |                        |   |
|--|--|--|--|--------------------------------------------------------------------------|------------------------|---|
|  |  |  |  | Innovation - AI<br>Innovation &<br>Sustainability                        | Ability<br>Enhancement | 1 |
|  |  |  |  | Research -<br>Research<br>Ethics<br>& Intellectual<br>Property<br>Rights | Ability<br>Enhancement | 1 |

| Third Year |                                        |                  |         | Fourth Year |                                                              |                     |         |
|------------|----------------------------------------|------------------|---------|-------------|--------------------------------------------------------------|---------------------|---------|
|            | Course Name                            | Course category  | Credits |             | Course Name                                                  | Course Category     | Credits |
| Semester-5 | Computer Vision                        | Program Core     | 3       | Semester-7  | Advanced AI Agents                                           | Program Core        | 3       |
|            | Applied Deep Learning                  | Program Elective | 4       |             | Responsible AI: Bias, Fairness and Safety in AI Applications | Program Core        | 3       |
|            | Generative AI: Application Engineering | Program Core     | 3       |             | AI Systems Engineering                                       | Program Core        | 3       |
|            | Natural Language Processing            | Program Core     | 4       |             | Internship / Industrial Training                             | Internship          | 6       |
|            | Machine Learning in Production         | Program Core     | 3       |             | Environmental Science and Management                         | Value Education     | 2       |
|            | Program Elective-I                     | Program Elective | 3       |             | Program Elective-IV                                          | Program Elective    | 3       |
|            | Institute / Industrial Summer Training | Internship       | 1       |             | Program Elective-V                                           | Program Elective    | 3       |
|            | AZ-700                                 | Certification    | 1       |             | Open Elective-I                                              | Open Electives      | 3       |
|            | RIE-3 (Semester-5)                     |                  |         |             | AZ-400                                                       | Certification       | 1       |
|            | Course Name                            | Course Category  | Credits |             | Professional Ethics                                          | Ability Enhancement | 1       |

|  |                                                     |                   |         |  |  |  |  |
|--|-----------------------------------------------------|-------------------|---------|--|--|--|--|
|  | Employability - Soft Skills-III                     | Skill Enhancement | 1       |  |  |  |  |
|  | Innovation - Finance & Economics Fundamentals       | Skill Enhancement | 1       |  |  |  |  |
|  | Research - Emerging Research Tools and Technologies | Skill Enhancement | 1       |  |  |  |  |
|  | RIE-4 (Semester-5)                                  |                   |         |  |  |  |  |
|  | Course Name                                         | Course Category   | Credits |  |  |  |  |

|                    |                                             |                        |                |                   |                                      |                        |                |
|--------------------|---------------------------------------------|------------------------|----------------|-------------------|--------------------------------------|------------------------|----------------|
|                    | Employability - Aptitude-III                | Ability Enhancement    | 1              |                   |                                      |                        |                |
|                    | Innovation Business Strategies & Operations | Ability Enhancement    | 1              |                   |                                      |                        |                |
|                    | Research - Statistical Data Analysis        | Ability Enhancement    | 1              |                   |                                      |                        |                |
|                    | <b>Course Name</b>                          | <b>Course Category</b> | <b>Credits</b> |                   | <b>Course Name</b>                   | <b>Course Category</b> | <b>Credits</b> |
| <b>Semester -6</b> | RAG & Vector Databases                      | Program Core           | 4              | <b>Semester-8</b> | Seminar on Current Trends in AI      | Program Core           | 1              |
|                    | Building LLMs                               | Program Core           | 4              |                   | Industry Integrated Capstone Project | Project Work           | 7              |
|                    | Agentic AI                                  | Program Core           | 3              |                   | Program Elective-VI                  | Program Elective       | 3              |
|                    | General Proficiency-II                      | Ability Enhancement    | 1              |                   | Open Elective-II                     | Open Electives         | 3              |

|                                      |                                               |                                                   |                     |  |                      |                     |   |
|--------------------------------------|-----------------------------------------------|---------------------------------------------------|---------------------|--|----------------------|---------------------|---|
|                                      | Program Elective-II                           | Program Elective                                  | 3                   |  | Research Methodology | Ability Enhancement | 2 |
|                                      | Program Elective-III                          | Program Elective                                  | 3                   |  |                      |                     |   |
|                                      | Introduction to SDGs                          | Value Education                                   | 1                   |  |                      |                     |   |
|                                      | DP-100                                        | Certification                                     | 2                   |  |                      |                     |   |
|                                      | RIE-5 (Semester-6)                            |                                                   |                     |  |                      |                     |   |
|                                      | Course Name                                   | Course Category                                   | Credits             |  |                      |                     |   |
|                                      | Employability - Soft Skills-IV                | Skill Enhancement                                 | 1                   |  |                      |                     |   |
|                                      | Innovation - Business Planning & Legal Basics | Skill Enhancement                                 | 1                   |  |                      |                     |   |
|                                      | Research - Domain specific Term paper         | Skill Enhancement                                 | 1                   |  |                      |                     |   |
|                                      | RIE-6 (Semester-6)                            |                                                   |                     |  |                      |                     |   |
|                                      | Course Name                                   | Course Category                                   | Credits             |  |                      |                     |   |
|                                      | Employability - Aptitude-IV                   | Ability Enhancement                               | 1                   |  |                      |                     |   |
|                                      |                                               |                                                   |                     |  |                      |                     |   |
|                                      |                                               | Innovation - Customer Engagement & Organic Growth | Ability Enhancement |  | 1                    |                     |   |
| Research - AI & Data-Driven Research |                                               | Ability Enhancement                               | 1                   |  |                      |                     |   |

## Elective Baskets

### Program Elective-I (Semester-5)

| Semester   | Course Name                            | Course Category  | Credits |
|------------|----------------------------------------|------------------|---------|
| Semester-5 | Knowledge Systems                      | Program Elective | 3       |
| Semester-5 | Conversational AI Systems              | Program Elective | 3       |
| Semester-5 | Advanced NLP and Language Intelligence | Program Elective | 3       |
| Semester-6 | Multimodal AI                          | Program Elective | 3       |
| Semester-6 | Advanced Deep Learning Architectures   | Program Elective | 3       |
| Semester-6 | Video Analytics and Edge Vision        | Program Elective | 3       |
| Semester-6 | Reinforcement Learning                 | Program Elective | 3       |
| Semester-6 | Intelligent Decision Systems           | Program Elective | 3       |
| Semester-6 | Autonomous and Multi-Agent Systems     | Program Elective | 3       |
| Semester-7 | Data Engineering for AI                | Program Elective | 3       |
| Semester-7 | MLOps and AI Deployment                | Program Elective | 3       |
| Semester-7 | Cloud and Edge AI Platforms            | Program Elective | 3       |
| Semester-7 | Time Series Analysis and Forecasting   | Program Elective | 3       |
| Semester-7 | Adversarial Machine Learning           | Program Elective | 3       |
| Semester-7 | Human-Centered AI and Compliance       | Program Elective | 3       |
| Semester-8 | AI Security                            | Program Elective | 3       |
| Semester-8 | LLM Fine-Tuning and Evaluation         | Program Elective | 3       |
| Semester-8 | AI Solution Architecture               | Program Elective | 3       |

*\*Students choose one course from each basket.*



4<sup>th</sup> Year Curriculum Design Framework [For Hons. / Hons. with Research Tracks of Four Year Undergraduate Programs (FYUP)]

| Extending Expertise | Course Category | Credits | Contextualizing Learning             | Course Category | Credits | Experiential Learning                  | Course Category | Credits |
|---------------------|-----------------|---------|--------------------------------------|-----------------|---------|----------------------------------------|-----------------|---------|
| DP-100              | Certification   | 2       | Open Elective-I                      | OE              | 3       | Social Internship                      | Internship      | 2       |
|                     |                 |         | Open Elective-II                     | OE              | 3       | Institute / Industrial Summer Training | Internship      | 1       |
|                     |                 |         | Environmental Science and Management | Value Education | 2       | Internship / Industrial Training       | Internship      | 6       |
|                     |                 |         | Professional Ethics                  | Value Education | 1       | Industry Integrated Capstone Project   | Project         | 7       |
|                     |                 |         | Introduction to SDGs                 | Value Education | 1       |                                        |                 |         |